

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 05 — BUILDING A SCADA HUMAN MACHINE INTERFACE

Student Edition

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Date	02-10-2026

Lab 05: Building a SCADA Human Machine Interface

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Apply basic high-performance HMI and situational-awareness principles.
- Create a tank, level indication, valve and process piping in FUXA.
- Connect the HMI to the OpenPLC tank-fill simulation.
- Bind HMI objects to PLC/Modbus tags and configure simple animation/state indication.
- Demonstrate that the HMI reflects the running process correctly.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password
SCADA VM	192.168.2.10	scada	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Note: Due to issues with FUXA the first part of the lab will not be completed

Step 1. Start the PLC and SCADA VMs in the isolated laboratory network. Start FUXA on the SCADA VM and open <http://127.0.0.1:1881/editor>.

Step 2. Create a New Project named with your registration number and LAB05, for example 2026XXXX_LAB05.

Step 3. Set a neutral gray background and enable the grid while arranging process graphics.

Step 4. From the Process Engineering objects, place a tank graphic. Add a bar gauge over/inside the tank and set its engineering range to 0–10 ft.

Step 5. Build a simple motor-actuated fill valve using the available valve/shape objects. Add a pipe from the valve to the tank. Keep the normal-state design visually simple.

Step 6. On the PLC VM, load/start the provided TankFillSim.st program in OpenPLC Runtime and confirm the status is Running.

Step 7. In FUXA, create or edit the Modbus TCP connection to the PLC at 192.168.1.10:502.

Step 8. Create/verify the Tank Level tag mapped to the OpenPLC input-register location used by the simulation (%IW80 / Input Register 80 in the teaching image).

Step 9. Create/verify the Tank Valve Open tag mapped to the OpenPLC discrete output used by the simulation (%QX0.0 / Coil 0 in the teaching image).

Step 10. Bind the bar gauge to the Tank Level tag. Configure Min = 0 and Max = 10.

Step 11. Bind the valve-state graphic to Tank Valve Open. Configure the normal/closed representation for value 0 and the open representation for value 1.

Step 12. Use a restrained high-performance style: neutral normal states, clear labels, and attention-drawing color only where it communicates a meaningful abnormal/alarm condition.

Step 13. Enable the PLC/FUXA connection and run the HMI. Observe the tank filling/draining and confirm the valve-state graphic changes with the process.

Step 14. Save/export the FUXA project and capture evidence of both the design view and running HMI.

Required Evidence & Submission

- Screenshot of the HMI in design/edit mode.
- Screenshot of the running HMI with a visible tank-level value.
- Screenshot of the configured PLC connection/tags with sensitive passwords excluded.
- The exported FUXA project file.
- A short design rationale explaining how the interface supports situational awareness.

Recommended filename: <RegistrationNo>_LAB05_Building_a_SCADA_Human_Machine_Interface.pdf plus the required project/capture file(s).

A short design rationale explaining how the interface supports situational awareness

This interface is explicitly engineered to minimize cognitive load and systematically build user Situational Awareness by mapping directly three distinct cognitive layers: Perception, Comprehension, and Projection. By structuring information to align with these mental processes, the system enables operators to instantly grasp real-time environmental data and confidently anticipate future states

SITUATIONAL AWARENESS		
LEVEL 1 Perception (What is it now?)	LEVEL 2 Comprehension (What does it mean?)	LEVEL 3 Projection (What's next?)

1. Level 1 SA: Perception (Acquiring Information)
The foundational tier focuses on helping the operator notice critical environmental elements without inducing data saturation.
2. Level 2 SA: Comprehension (Making Sense of Information)
Knowing a metric exists is insufficient; operators must easily diagnose its significance.
3. Level 3 SA: Projection (Anticipating Future Events)
The ultimate benchmark of an advanced interface is enabling predictive decision-making.

Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

Response: P&IDs are the drawings used by engineers to design and build an industrial process. Typically, they will show the instrumentation associated with a process as well as information about the indications present in a process. P&ID Influences HMI Layout in several key ways

- ✓ Defining Process Flow and Visual Hierarchy
- ✓ Tagging, Nomenclature, and Alarm Systems
- ✓ Control Loop Grouping and Interconnectivity
- ✓ Component Inventory and Asset Mapping

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Response: High-performance HMIs avoid unnecessary animation and excessive color to minimize operator cognitive load and dramatically improve situational awareness. Experts around human perceptual psychology, dictate that an interface should remain visually calm until something genuinely requires immediate human intervention. Excessive color is avoided due to the following

- ✓ Dilution of Alarm Significance
- ✓ Reduced Visual Fatigue
- ✓ Elimination of Environmental Glare & Illusions

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Response: In SCADA and PLC automation systems, a Tank Level tag and a Tank Valve Open tag serve entirely different roles in monitoring and control. They differ fundamentally in both data type and operational behavior:

Attribute	Tank Level Tag	Tank Valve Open Tag
Primary Data Type	Analog (Float / Real / Integer)	Discrete / Digital (Boolean / Bit)
Values	Continuous range (e.g., 0.0 to 100.0%, or 0 to 4095 raw)	Binary states (0 / 1, False / True, OFF / ON)
Physical Component	Level Transmitter	Limit switch or valve actuator
Behaviour	Continuous fluctuation based on material flow	State-based transitions (instantaneous changes)

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

Response: When a HMI displays stale or incorrect PLC data, the consequences can lead to negative consequences in industrial operation. Because operators rely on the HMI as their "eyes and ears," making decisions based on data that looks valid but is actually outdated or wrong can lead to serious operational failures.

Here is what can happen operationally, categorized by the level of risk:

- ✓ Equipment Damage & Mechanical Failures
- ✓ Quality Escapes & Material Waste
- ✓ Production Blind Spots & Blind Flying
- ✓ Operational Downtime & Safety Risks

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.
Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.		